

WASTEWISE-ML · REQUIREMENTS

WasteWise-ml requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise-ml requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see WML-PRB-001). No model has been trained, so every performance requirement is unverified; the status column gives the concept's expected position and says plainly where it is not met.

The **reference use case** is one person at a sorting bench or on the floor, holding a low-cost Android phone (or using it on a stand) over one item at a time, on a dry mixed stream, in daylight or a lit shed, with no network.

Table 1. Requirements, targets and status at TRL 2.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Material class accuracy	Top-1 accuracy of 90 % or more on the seven classes in <code>ml/data/taxonomy.yaml</code> , on items the model does not abstain on, on the field evaluation set (R11)	Evaluation on the held-out field set	Unverified; at risk. TrashNet's authors reported about 75 % on a clean six-class set
R2	Plastic grade accuracy from a photo	90 % or more for PET bottles and trays, 80 % or more for other rigid HDPE and PP, on non-abstained items	Field set, ground truth from NIR (WasteWise Scan) or legible resin codes	Not met by design for opaque HDPE versus PP and for black plastics; these go to "Unsure" and to WasteWise Scan
R3	Hazard flag	Recall of 98 % or more for batteries, sharps and chemical containers; a hazard result is never offered as a sortable grade	Field set with at least 200 hazard items	Unverified; at risk because hazards are rare in training data
R4	Abstain when unsure	Model says "Unsure" below a set confidence; coverage (share of items answered) 70 % or more while meeting R1	Coverage and accuracy curve on the field set	Unverified; about 75 % coverage assumed (estimate)
R5	Offline, on-device speed	Result within 0.3 s of the photo on the reference phone (R12), with no network	Timing on the reference phone	Met on paper: about 0.05 to 0.2 s for MobileNetV3-Large in 8-bit form (estimate)
R6	Small download	Model file 10 MB or less; app with model 30 MB or less	File sizes of the exported models	Met on paper: about 5 to 6 MB (estimate)
R7	Quick to use	3 s or less from placing an item to seeing the result, including tap and camera focus	Timed trials with users	Unverified
R8	Usable without reading	Result shown as icon, color and bin number, with optional voice in the local language; users new to the app sort correctly after a 15 min briefing	Co-design sessions and usability trials	Unverified; depends on co-design
R9	Local grades	Buyer grade names and bin numbers set per site in a configuration file, without retraining, for grades the model already separates	Configuration review	Met on paper by the grade mapping layer; new visual grades still need retraining
R10	Battery life	A full 8 h shift with up to 500 scans uses 30 % or less of a 4,000 mAh phone battery	Energy estimate; later measurement	Met on paper with tap-to-scan (about 7 % estimate); not met with a live camera preview all shift
R11	Field evaluation set	2,000 or more verified images from at least two sites, 50 or more per grade, never used for training	Dataset manifest	Not met: the set does not exist yet
R12	Reference phone	Runs on an Android phone costing about \$150 or less, 3 GB RAM, 2019 or later chipset	Device test	Unverified; assumed from published latency

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R13	Open and licensed	Code and weights MIT; every dataset license in <code>ml/data/SOURCES.md</code> allows training and publishing weights; a model card with per-class results and known failure modes	License review	Partly met: candidate sources listed, licenses still to confirm
R14	Consent and privacy	Field photos only with informed consent; no faces or identifiable people kept; upload opt-in; pickers' organization co-owns field data (proposed)	Data protocol review with the partner	Unverified; protocol to be written with the partner

■ Requirements not met or at risk

- **R2 is not met by a camera alone** for opaque HDPE versus PP and for black plastics. The concept routes these to "Unsure" and to WasteWise Scan.
- **R11 is not met:** there is no field evaluation set yet, so R1 to R4 cannot be verified.
- **R10 is not met** if the app runs a live preview all shift; the concept therefore uses tap-to-scan.
- **R1, R3 and R4 are at risk** because public data is small, clean and not organized by grade.

■ Assumptions

- Seven material classes from the taxonomy (plastic, paper, metal, glass, organic, composite and hazardous), with about 22 grades below them.
- A phone battery of 4,000 mAh at 3.85 V holds about 15 Wh; screen and camera draw about 2.5 W while scanning (estimate).
- Published MobileNetV3 speeds on a Pixel 1 large core are a fair lower bound; a low-cost 2019 or later phone is taken as 1 to 4 times slower (estimate).
- About 75 % of items are answered confidently, 22 % are "Unsure" and 3 % are hazard flags in a dry mixed stream (estimates for the flow diagram, to be replaced by field data).